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Revisiting health state preferences after 20 years: A new EQ-5D-3L value set for the Netherlands

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Abstract

Introduction Value sets for the EQ-5D-3L have been developed since the 1990's, using methods that are now considered outdated. The Dutch EQ-5D-3L value set was developed using data collected in 2003, and population's preferences may have shifted over time. Consequently, the existing value set may no longer accurately represent the preferences of the Dutch general population. This study aims to develop a new EQ-5D-3L value set using state-of-the-art methods and compare it with the existing value set.

Methods A nationally representative sample of 417 Dutch adults, stratified by age and sex, completed 12 composite time trade-off (cTTO) tasks via online EuroQol Valuation Technology interviews. The data were modelled using a Tobit model that accounts for heteroskedasticity and the censored nature of cTTO data. Agreement between the new and existing value set was assessed using Bland-Altman and scatter plots.

Results Pain/discomfort and mobility were the most important dimensions in the new value set, whereas pain/discomfort and anxiety/depression were most important in the existing value set. The lowest health state value was −0.723, considerably lower than the −0.329 of the existing value set. The Bland-Altman and scatter plots indicated limited agreement between the two value sets.

Conclusion The new Dutch EQ-5D-3L value set differs substantially from the existing value set. We recommend its adoption and replacement of the previous value set. Our findings suggest that other countries with older value sets should consider similar updates to ensure accurate representation of contemporary societal preferences.

Keywords EQ-5D-3L · Value set · Value set update · Value set redundancy · Netherlands

Key points

Health utilities for the EQ-5D-3L for the Netherlands have changed over 20 years. These changes could be caused by improvements measurement and modelling methods, changes in preferences of the population, as well as changes to the demographic composition of the population of the Netherlands. We recommend to replace the existing EQ-5D-3L value set for the Netherlands with the currently developed one, and other countries to consider similar updates for older value sets.

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Introduction

The EQ-5D-3L is among the most widely used measures of health-related quality of life (HRQoL). It describes health states across five dimensions (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression), with three levels of severity per dimension (no problems, moderate problems, severe problems) [1]. This 5-dimension, 3-level structure enables the distinction of 243 unique health states, denoted by 5-digit codes representing the level of problems on each dimension. For instance, health state 12313 indicates no problems with mobility (level 1),

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some problems with self-care (level 2), inability to perform usual activities (level 3), no pain/discomfort (level 1), and extreme anxiety or depression (level 3).

To enable the calculation of Quality Adjusted Life Years (QALYs), country-specific value sets have been developed. These sets assign quality weights to each EQ-5D-3L health state based on preference-based trade-offs between quality and duration of life, anchored on a health-utility scale where 0 represents immediate death and 1 represents full health. Values below 0 indicate health states considered worse than death (WTD). For the Netherlands, such a value set has been available since 2006 [2], which has been extensively employed in health economic evaluations and is among the recommended instruments for cost-utility analyses in the country [3].

However, the current Dutch EQ-5D-3L value set, published 18 years ago and based on data collected 21 years ago, may no longer accurately reflect contemporary Dutch population preferences. The Netherlands has undergone significant demographic changes during this period, and societal preferences may have evolved independently of these demographic shifts. Furthermore, methodologies for eliciting health-state preferences have substantially advanced since the current value set's publication.

A notable methodological advancement has been the EuroQol Group's development of the standardized EuroQol Valuation Technology (EQ-VT) protocol, which represents a significant improvement over the previously used Measurement and Valuation of Health (MVH) protocol [4–6]. While the MVH protocol used standard Time Trade-Off (TTO) with a lower bound of −39 for negative values—often arbitrarily transformed to a lower bound of −1—the EQ-VT protocol is based on the composite Time Trade-Off (cTTO) method. This method incorporates lead-time TTO (LT-TTO) for eliciting negative values, naturally imposing a lower bound at −1 and avoiding arbitrary transformations [7, 8]. Additionally, the EQ-VT protocol includes comprehensive quality control procedures to ensure data quality, addressing concerns raised about the impact of data quality on valuation outcomes [9].

Given these methodological advances and the potential obsolescence of older value sets, this study aims to develop a new EQ-5D-3L value set for the Netherlands using state-of-the-art methods. By comparing this new value set with the existing one, we seek to assess the extent of changes in societal preferences and the impact of improved methodologies on health state valuations [10, 11]. This comparison may provide insights into changes in health state preferences in the Netherlands, and potentially contribute to the broader discussion on the need for updating older value sets in other countries to ensure they accurately reflect current societal preferences for health-related quality of life. Potentially updating the EQ-5D-3L value set is relevant, as it is still recommended as the next-best alternative in cost-utility analyses for ZIN in the Netherlands, with the reference case being EQ-5D-5L [3].

Methods

Overview

An EQ-VT study was executed using a representative sample of the Dutch population in terms of age and sex. Respondents were recruited by a marketing research company and interviewed by trained interviewers. Quality control measures were implemented to ensure data integrity, and regressions methods used to analyze the data and produce the new EQ-5D-3L value set. Finally, the new value set was compared with the previous one.

Health state valuations

In our study, health states were valued using cTTO. For health states that respondents perceive as better than death, cTTO presents a conventional TTO task in which participants make adaptive choices between a duration of X years in full health and 10 years in the health state to be valued. Through an iterative process, X is adjusted to identify the point of indifference, where the health state value is calculated as X/10. For health states perceived as worse than death, a lead-time TTO is applied. In this case, respondents choose between X years in full health and a fixed duration consisting of 10 years in full health followed by 10 years in the specified health state. Here, X is again varied until a point of indifference is reached, with the health state value now determined by (X − 10)/10. This approach enables cTTO to generate health state values that range from − 1 to 1 [5, 6].

Experimental design

Each respondent completed 12 cTTO tasks, comprising a valuation of the worst possible health state, 33333, along with 11 additional health states selected from a subset of 196 of the 243 possible EQ-5D-3L health states. This set was derived by removing certain states from the full set of 243 EQ-5D-3L health states: 11111 (full health), 33333 (worst health, already included), and 45 health states that were deemed implausible. Implausibility was defined as the presence of level 3 on mobility ("confined to bed") paired with level 1 ("no problems") on either self-care or usual activities, following Viney et al. [12]. The remaining 196 states were scored using the Lamers et al. Dutch EQ-5D-3L value set, and bracketed into 11 severity groups based on their utility score [2]. Respondents were randomly assigned one health state from each severity block, which in combination with state 33333 totals 12 cTTO respondents per respondent, stratified by health state severity. This experimental design ensured broad representation of the EQ-5D-3L system while excluding logically inconsistent states.

Recruitment, sampling and survey administration

Bureau Fris, a Dutch market research company, recruited participants from their existing panel to achieve a sample representative of the Dutch general population by age and sex. The target sample size was 400 participants, with participants receiving a 30-euro compensation for their time. Respondents were quota sampled on age and sex, while efforts were made to collect a sample which closely represented the Dutch population in terms of geographical region and level of education as well. The interviews were conducted remotely by four trained interviewers, using Zoom videoconferencing software, a validated approach shown to produce results comparable to face-to-face interviews [13, 14].

Survey structure

The survey began with respondents providing informed consent. This was followed by a set of questions about the demographic characteristics of the respondents, which included age, sex, highest education attained, and province of residence. Next, respondents self-completed the EQ-5D-3L instrument.

Following this, respondents proceeded to the cTTO valuation tasks. They first completed two practice questions to become familiar with the cTTO task and its procedures. In these practice tasks, respondents valued “being in a wheelchair” and either “a health state much better than being in a wheelchair” or “a health state much worse than being in a wheelchair”, depending on whether the first question ended in the lead-time TTO or the regular TTO.

Respondents then completed three additional practice EQ-5D-3L questions (21112, 32323 and 13311) to further orient themselves to the cTTO process. After the practice rounds, each respondent valued 12 EQ-5D-3L health states using cTTO. Upon completing all the cTTO tasks, respondents were presented the rank order of their responses and asked to flag any they thought were misordered using the feedback module [15].

Finally, respondents answered a short questionnaire about their preferences regarding different durations, after which the interview was concluded and they were thanked for their participation.

Quality control

Several quality control measures were implemented to ensure data integrity. The interviewers received training from the principal investigators and conducted 5 practice interviews with acquaintances, which were not included in the final sample.

After each batch of 5–10 interviews collected by an interviewer, the data were imported and inspected for protocol compliance and interviewer effects. Interviews were flagged as potentially problematic if: (1) respondents did not enter the lead-time TTO in either of the two wheelchair example tasks, suggesting they did not receive a full explanation of that method; (2) respondents spent less than 3 min total on the two wheelchair example questions, making it unlikely the cTTO task was properly explained; (3) respondents completed the 12 actual cTTO questions in less than 6 min; or (4) health state 33333 did not receive the lowest value and another state received a value at least 0.5 lower [10].

If more than 40% of an interviewer’s interviews were flagged, their data was removed and the interviewer was retrained. Interviewer effects were also examined by comparing response distributions across interviewers. Respondents were excluded from the final sample if regression analysis showed a positive slope for the level sum score, indicating they viewed more severe health states as better than less severe ones.

Statistical analyses

The cTTO data were analyzed using a heteroskedastic Tobit model with the utility function as described by Eq. (1):

$$
\begin{aligned}
U_j = \\
\beta_0 + \beta_1 MO2_j + \beta_2 MO3_j + \beta_3 SC2_j + \\
\beta_4 SC3_j + \beta_5 UA2_j + \beta_6 UA3_j + \beta_7 PD2_j + \\
\beta_8 PD3_j + \beta_9 AD2_j + \beta_{10} AD3_j + \epsilon_j
\end{aligned}
\tag{1}
$$

Here, U_j represents the utility assigned to health state j, while the variables $MO2_j$, $MO3_j$,…., $AD3_j$ are dummy variables indicating the presence of the respective level of problems for each dimension. For example, if health state j is 12,313, then $SC2_j$, $UA3_j$ and $AD3_j$ equal 1, and all other variables in the model will equal zero. The β parameters are the estimated quality weights associated to the level-dimension dummies and β_0 is the intercept. ϵ_j is the error term, whose variance is assumed to depend on the independent variables, as shown in Eq. (2):

Table 1 Demographic characteristics of the sample

	cTTO sample		Netherlands (CBS 2024)
Age	**N**	**%**	**%**
18–24	42	10.1%	11.0%
25–34	67	16.1%	16.3%
35–44	68	16.3%	15.1%
45–54	74	17.7%	15.7%
55–64	70	16.8%	16.8%
65+	96	23.0%	25.1%
Sex	**N**	**%**	**%**
Male	208	49.9%	49.70%
Female	207	49.6%	50.30%
Other	2	0.5%	< 1%
Education	**N**	**%**	**%**
Low	31	7.4%	26.6%
Middle	173	41.5%	37.0%
High	213	51.1%	36.4%
Region	**N**	**%**	**%**
North	34	8.1%	9.80%
East	67	16.1%	18.60%
Central	57	13.7%	10.30%
West	170	41.1%	38.00%
South	89	21.3%	23.30%

Regions: North (Drenthe, Friesland and Groningen), East (Gelderland and Overijssel), Central (Flevoland and Utrecht), West (Zuid-Holland and Noord-Holland), South (Zeeland, Noord-Brabant and Limburg)

$$\sigma_j^2 = \exp(\gamma_0 + \gamma_1 MO2_j + \gamma_2 MO3_j + \gamma_3 SC2_j + \gamma_4 SC3_j + \gamma_5 UA2_j + \gamma_6 UA3_j + \gamma_7 PD2_j + \gamma_8 PD3_j + \gamma_9 AD2_j + \gamma_{10} AD3_j) \quad (2)$$

Additionally, it is assumed that values of −1, the lowest possible value elicited in the cTTO task, are censored. This is described in Eq. 3:

$$U = \begin{cases} U" \, if \, U" > -1 \\ -1 \, if \, U" \leq -1 \end{cases} \quad (3)$$

Here U represents the cTTO responses and $U"$ the underlying latent value. After estimating the Tobit model, the resulting utility function can be used to predict values for all 243 EQ-5D-3L health states, with Eq. 3 linking the predicted latent utilities to the censored cTTO responses at the −1 lower bound.

To compare the new value set to the previous one, scatterplots, Bland-Altman plots, and summary statistics were used, including mean difference, mean absolute error (MAE), root mean square error (RMSE), and Lin's Concordance Correlation Coefficient (CCC) [16]. – [17] These analyses provide a comprehensive assessment of the agreement between the two value sets.

Table 2 Estimated value set

	Coef	SE	*p*-value
mo2	−0.045	0.021	0.029
mo3	−0.424	0.032	0.000
sc2	−0.059	0.024	0.015
sc3	−0.144	0.024	0.000
ua2	−0.011	0.024	0.662
ua3	−0.132	0.025	0.000
pd2	−0.114	0.022	0.000
pd3	−0.509	0.025	0.000
ad2	−0.096	0.023	0.000
ad3	−0.360	0.024	0.000
constant	0.846	0.030	0.000
v(33333)	−0.723		

SD standard deviation, V(33333); the estimated value for health state 33333

Results

Data collection was completed during the summer of 2024, with four trained interviewers conducting a total of 417 interviews among the Dutch general population. Quality control measures were met by all interviewers, and thus no data were excluded due to interviewer-related protocol violations. Table 1 summarizes the demographic characteristics of the sample, along with comparisons to Dutch population statistics [18, 19]. The sample was representative in terms of age, sex, and region; however, individuals with lower educational levels were underrepresented. In total, six respondents were removed due to a positive regression slope for the level sum score, resulting in a final sample of 411 respondents for analysis.

Model estimates

The heteroskedastic Tobit model results are displayed in Table 2, with dummy variables mo2 to ad3 capturing the reduction in health utility values associated with each level of problem in the respective dimensions. For instance, mo2 reflects the disutility of experiencing some mobility issues, while pd3 represents extreme pain or discomfort. The constant term suggests that moving from full health (state 11111) to any other state incurs a baseline reduction in utility, independent of the specific dimensions affected.

Among the dimensions, pain/discomfort had the highest impact, with a level 3 weight of −0.509, followed by mobility (−0.424), anxiety/depression (−0.360), self-care (−0.144), and usual activities (−0.132). The constant term of 0.846 indicated an additional reduction of 0.154 associated with moving away from full health. The resulting value set is represented in Eq. 4:

$$v\,(HS) = 1 - 0.154 * N1 - 0.045 * mo2 - 0.424 * mo3 - 0.059 * sc2 - 0.144 * sc3 - 0.011 * ua2 - 0.132 * ua3 - 0.114 * pd2 - 0.509 * pd3 - 0.096 * ad2 - 0.360 * ad3 \tag{4}$$

where $v\,(HS)$ is the value of a given health state based on the associated dummy variables (e.g., mo2, sc2) and $N1$, which denotes the move from state 11111 to a health state with health problems. For example, health state 12313 has a value of:

$$v\,(12312) = 1 - 0.154 * N1 - 0.059 * sc2 - 0.132 * ua3 - 0.360 * ad3 = 0.295 \tag{5}$$

An overview of all 243 unique EQ-5D-3L values derived using this algorithm is provided in the electronic supplementary materials, Table A2

Comparison with previous value set

Figure 1 presents a scatter plot comparing the utility values derived in this study with those from the previous Dutch value set. The plot shows an alignment along the 45-degree line for moderate health states, suggesting general concordance in utility values. However, larger discrepancies are evident for more severe health states, particularly those with negative values, indicating scale differences between the two sets.

The Bland-Altman plot in Fig. 2 further illustrates these discrepancies. In this plot, the differences between the two value sets are plotted against the average value for each health state. The plot reveals that differences are more pronounced for health states rated below zero, with the new value set tending to yield lower values for these states. In contrast, values for moderate health states are generally higher in the new set compared to the previous one.

Summary statistics reinforce these findings. The mean difference between the new and previous value sets was 0.049, with a mean absolute error (MAE) of 0.116 and a root mean square error (RMSE) of 0.144. Lin's Concordance Correlation Coefficient (CCC) was calculated at 0.890, indicating poor agreement cf. McBride [20].

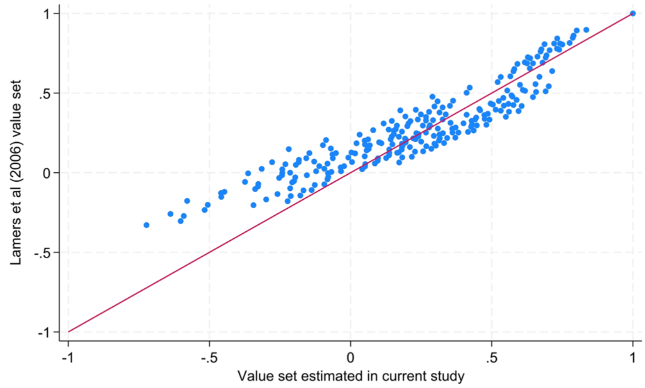

Fig. 1 Scatter plot Lamers et al. (2006) estimates versus current study estimates

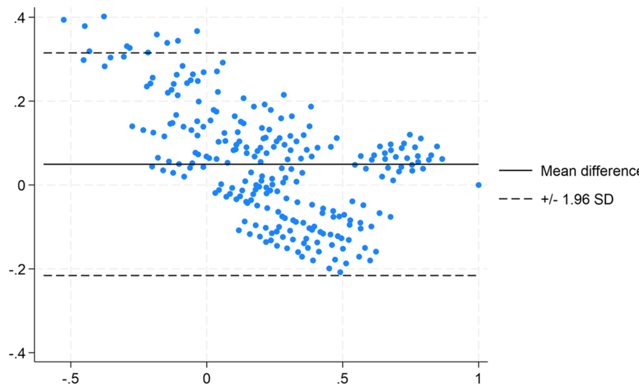

Fig. 2 Bland-Altman plot Lamers et al. (2006) estimates versus current study estimates

Discussion

This study provides an updated EQ-5D-3L value set for the Netherlands, capturing the current health preferences of the Dutch adult population. In this new value set, pain/discomfort is the most important dimension, followed by mobility, anxiety/depression, self-care, and usual activities. Values range from − 0.723 for the worst health state (33333) to 1 for full health (11111). Comparisons with the existing Dutch EQ-5D-3L value set [2], which had a range of −0.329 to 1, reveal poor agreement. The current study has resulted in a value set with slightly lower values for mild health states, slightly higher values for moderate states, and substantially lower values for severe states. Additionally, the relative importance of dimensions differs: while anxiety/depression is the second most important dimension after pain/discomfort in the previous value set, mobility holds this position in the present study. When compared to the Dutch EQ-5D-5L value set, there are also substantial differences, most notably in the ordering of dimension importance [21]. With pain/discomfort and anxiety/depression being the most important dimensions, and mobility receiving a much smaller weight. However, the level descriptors of the most extreme response options differ substantially between the EQ-5D-3L and EQ-5D-5L, with the EQ-5D-3L describing this level as "Confined to bed", whereas it is described as "Unable to walk about" in the EQ-5D-5L.

These differences highlight that value sets can evolve over time. Methodological advancements in this study offer a potential explanation for the observed shift in health state values. First, the cTTO approach used here avoids the arbitrary transformations for negative values required by earlier methods. Second, incorporating quality control during TTO data collection likely impacts response distributions, as prior research shows that rigorous quality control can significantly affect valuation outcomes [10]. Third, the modelling approach, which accounts for data censoring and heteroskedasticity, provides a better fit for the data compared to earlier models, although it can expand the scale length. For instance, studies employing Tobit models, such as the Dutch EQ-5D-5L valuation study [21], reveal substantial discrepancies in scale length (up to 0.093) compared to simpler random-intercept models like that used in the Lamers et al. study. Beyond methodological improvements, preference shifts and demographic change in the Netherlands may also have influenced the observed value set differences, although demographic change was in a recent study found to be incapable of exerting substantial value set change in a time span of 20–25 years [11]. Still, a substantial shift in the relative importance of the EQ-5D-3L domains was observed in the current study, as well as a large shift in the length of the scale. While changes in scale could potentially be explained by differences in measurement and modelling methods, it is difficult to make inferences about the cause of changes in the relative importance of dimensions.

The differences found between the existing Dutch EQ-5D-3L value set and the currently derived values has two implications for decision making: (1) the relative importance of dimensions has changed between the two value sets, which shifts the relative importance from anxiety/depression more towards mobility, which may have an impact on the Incremental Cost Effectiveness Ratios (ICERs) of interventions (2) QALY gains are larger for moving from severe health states to moderate or mild health states, which could reduce the cost-per-QALY for interventions that facilitate such changes in health status, making ICERs based on EQ-5D-3L utilities more favourable for such interventions. To a lesser degree, the opposite may be true when moving from moderate health problems to mild problems or full health, as the new value set tends to predict higher values for moderate states as compared to Lamers et al., while the values for mild states are somewhat lower. However, in the case of the Netherlands, the impact on decision making may be limited, as Zorg Instituut Nederland recommends EQ-5D-5L as the primary outcome used in cost-utility analysis in their guidelines, with EQ-5D-3L being an alternative, although it is used less frequently [21, 22]. However, in other countries in which only an older value set is available, value set updates such as the current one may be much more influential on policy terms. Countries in which an older EQ-5D-3L value set exist, but do not yet have an EQ-5D-5L value set may consider collecting valuation data for the EQ-5D-3L as part of their EQ-5D-5L valuation study, as done in e.g. Hungary and Romania [23–25].

A limitation of this study is that the study's research design does not allow us to disentangle the relative contributions of improved methods versus preference shifts and demographic changes. In addition, even though the collected sample is representative in terms of age, sex, and region, respondents with lower education were underrepresented. Furthermore, our data were collected via videoconferencing interviews. Although the outcomes of such interviews have been shown to be equivalent to face-to-face interviews, there may be a sample selection effect, with some respondents unwilling to participate in videoconferencing interviews [13, 14]. Finally, even though a state-of-the-art modelling approach was used, further refinements may be possible, such as the accommodation of preference heterogeneity or inclusion of interaction terms. Such refinements were beyond the scope of this study but remain promising areas for future research.

In conclusion, the presented Dutch EQ-5D-3L value set better reflects current societal preferences for health-related quality of life. Given the methodological advancements and improved data quality control procedures, we recommend adopting this new value set to replace the previous one, while acknowledging that the most recent guidelines for cost-utility analyses in the Netherlands recommend the use of the EQ-5D-5L instead of the EQ-5D-3L instrument [25]. Accordingly, our findings primarily underscore the importance of periodic value set updates in other countries to ensure alignment with contemporary preferences, ultimately supporting more accurate and relevant economic evaluations in healthcare. This could for example be done by monitoring the stability of preferences and periodically testing whether there are any differences that one might require an update to a value set. An example of how this could be implemented can be read in Jensen et al., who found preferences to be stable in Denmark over a 5-year period [26].

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10198-025-01892-2.

Author contributions Both authors met the ICMJE authorship criteria. BR and MFJ were jointly responsible for the study design. BR was responsible for the data collection, and conducted the data analysis. The results were interpreted together by BR and MFJ. BR drafted the first version of the manuscript, while MFJ reviewed and revised it for important intellectual content. BR and MFJ both have read and approved the final version of the manuscript. BR and MFJ agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Data availability The data that was collected in the study is available from the corresponding author upon reasonable request.

Declarations

Ethics approval Ethical approval was obtained from Erasmus School for Policy and Management's Research Ethics Review Committee, reference number ETH2324-0619. We certify that the study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards.

Informed consent Respondents were provided with an information sheet and completed a consent form before participating in this study.

Competing interests BR and MFJ are members of the EuroQol Group. BR is employed by the EuroQol Research Foundation.

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